

This can take a little getting used to. I have caused many bugs by forgetting that the base behavior looks for an exact match or sticking an asterisk into the string intending a "starts-with" semantic. Field look-ups can feel a little “anti-Pythonic”: it is just weird to change the name of function’s argument to change the behavior of the function. With practice, though, you’ll get used to it.

There are many field look-up modifiers. All value comparison is done using field look-ups: greater than, less than, greater than equal, and so on. Although it might feel more natural to specify *J\**, that syntax won’t work for nonstring values. Consider the queries *after this date* and *smaller than this number*, you just can’t stick a star on the end of those. The field look-up mechanism works regardless of the type of data being queried. For example, to find all musicians born in 1945 or later, you use `__gte` to find dates greater than or equal to a Python date object containing January 1, 1945:

```
>>> Musician.objects.filter(birth__gte=date(1945, 1, 1))
<QuerySet [<Musician: Musician(id=1, last_name=Vai)>, <Musician:
➡ Musician(id=3, last_name=Bonham)>]>
```

Some of the more commonly used field look-ups are in table 4.1.

**Table 4.1** Commonly used field look-ups

| Field Look-up              | Name                         | Description                                                                                                                                                        |
|----------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>__contains</code>    | Contains                     | Text field contains the given value                                                                                                                                |
| <code>__icontains</code>   | Case-insensitive contains    | Text field contains the given value, ignoring case                                                                                                                 |
| <code>__gt</code>          | Greater than                 | Objects with a field greater than the given value                                                                                                                  |
| <code>__gte</code>         | Greater than equal           | Objects with a field greater than or equal to the given value                                                                                                      |
| <code>__in</code>          | In iterable                  | Objects with a field value matching one of the values given in an iterable. Example: <code>id__in=[1, 4, 6]</code> matches objects with an ID field of 1, 4, or 6. |
| <code>__lt</code>          | Lesser than                  | Objects with a field lesser than the given value                                                                                                                   |
| <code>__lte</code>         | Lesser than equal            | Objects with a field lesser than or equal to the given value                                                                                                       |
| <code>__startswith</code>  | Starts with                  | Text field starts with the given value                                                                                                                             |
| <code>__istartswith</code> | Case-insensitive starts with | Text field starts with the given value, ignoring case                                                                                                              |
| <code>__endswith</code>    | Ends with                    | Text field ends with the given value                                                                                                                               |
| <code>__iendswith</code>   | Case-insensitive ends with   | Text field ends with the given value, ignoring case                                                                                                                |